

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE/IT)

May 2014

IT402 WIRELESS COMMUNICATION AND MOBILE COMPUTING

Date: 21.05.2014, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions.

[11]

- [A] What is super high frequency? [1]
- [B] What is delay spread in multipath propagation? [1]
- [C] _____ is an effect caused by multipath propagation. [1]
(a) Delay Spread (b) Spectrum Distribution (c) path improvement (d) Sectorizing
- [D] What is use of TMSI and MSRN numbers? [1]
- [E] Explain 802.11. [7]

Q-2 Answer the following questions.

[12]

- [A] Explain authentication mechanism used in GSM. [3]
- [B] Explain types of hotspot. [4]
- [C] What are the drawbacks of Cellular System (GSM)? [5]

OR

- [A] What are the features of CDMA over GSM network? [3]
- [B] What are the sequence of steps followed by Mobile Originated Call and Mobile Terminated Call? [4]
- [C] What is the roll of Modulation/Multiplexing in Wireless Communication? [5]

Q-3 Answer the following questions.(Attempt any Two)

[12]

- [A] Explain TDMA.
- [B] What is Hidden and exposed Terminal?
- [C] Discuss GPRS architecture.

SECTION-II

Q-4 Answer the following questions.

[11]

- [A] Define Radio Handover. [1]
- [B] Define FDMA. [1]
- [C] What is Minimal IP-in-IP encapsulation? [1]
- [D] Draw and explain architecture of GSM. [8]

Q-5 Answer the following questions.

[12]

- [A] Explain Link Manager Protocol In Bluetooth. [3]
- [B] Explain routing for wired and wireless networks. [4]
- [C] Write a short note on HiperLAN. [5]

OR

- [A] Explain in brief Satellite Network.
- [B] Explain BLUETOOTH protocol stack.
- [C] Explain indirect TCP.

[3]
[4]
[5]

Q-6 Answer the following questions.(Attempt any Two)

[12]

- [A] Write down the advantages and disadvantages of Snooped TCP.
- [B] Discuss in brief Mobile IP.
- [C] Explain Inter frame spaces in the context of Wireless Lan.